axion emissivity via pionic Compton-like processes.

In this work, we follow the detailed calculation of Refs. [41, 74] to implement the emission of MeV-scale ALPs via nuclear processes. As discussed in Refs. [41, 42], under the assumption of a sufficiently high pion fraction in the SN core, the ALP emission spectrum in the free-streaming regime $g_{aN} < 10^{-8}$ is characterized by a bimodal shape. The ALP spectrum induced by nuclear couplings observed at large distance from the source is displayed in Fig. 2, where the different colors refer to different ALP masses. In particular, all the spectra are integrated over the SN volume and over the duration of the SN cooling phase $\Delta t \sim 10$ s. NN bremsstrahlung emission spectrum (solid lines) is characterized by a quasi-thermal spectral shape peaking at energies $E_a \sim m_a + 50\,\mathrm{MeV}$, while the pion conversion contribution (dashed lines) shows a peak around $E_a \sim m_\pi + \frac{3}{2}T \sim 200\,\mathrm{MeV}$ [41–43]. We can observe that for $m_a \gtrsim 3T \sim 90\,\mathrm{MeV}$ bremsstrahlung is suppressed compared to pionic processes, which become dominant for ALP masses $m_a \gtrsim 30\,\mathrm{MeV}$. The ALP production spectra can be plugged into Eq. (21) to determine the associated positron spectra. In this context, ALPs produced in the SN core via nuclear processes decay in electron-positron pairs $a \to e^+e^-$ as argued above, yielding a positron injection in the ISM. We highlight that, distinctly from the previous section, in this scenario ALP production and ALP decays depend on different unrelated couplings.

Nevertheless, Ref. [44] has pointed out that ALPs coupled to nuclear matter are naturally provided with a QCD-induced photon coupling, which can be estimated as

$$g_{a\gamma} \simeq \frac{\alpha_{em}}{2\pi m_N} \times \Big[-\frac{1.92}{1.59\,c_d - 0.52}\,c_g - \left(\frac{0.71}{1.59\,c_d - 0.52}\,c_g + 0.79 \right) \frac{m_a^2}{m_\pi^2 - m_a^2} \Big] \tag{9}$$

where $m_\pi = 135\,\mathrm{MeV}$ is the pion mass, $\alpha_{em} = 1/137$, while c_g and c_d are model-dependent constants. As a benchmark case, here we set $c_g = 1$, $c_d = 0$. The presence of a non-vanishing photon coupling allows MeV-scale ALPs to efficiently decay into photon pairs, opening an additional decay channel which is competitive with electronic decays for ALP-electron couplings $g_{ae} \lesssim 10^{-16}$. Therefore, in this scenario, the total ALP decay length is given by

$$\lambda_{\mathrm{ALP}} = \left(\lambda_{\mathrm{ALP},e}^{-1} + \lambda_{\mathrm{ALP},\gamma}^{-1} \right)^{-1} , \tag{10}$$

where $\lambda_{\mathrm{ALP},e}$ and $\lambda_{\mathrm{ALP},\gamma}$ are the ALP decay lengths through electronic and photon decays, respectively (see Refs. [27, 75] for their explicit expression). Thus, the induced positron spectra have to be rescaled by the branching ratio for ALP decays into electron-positron pairs

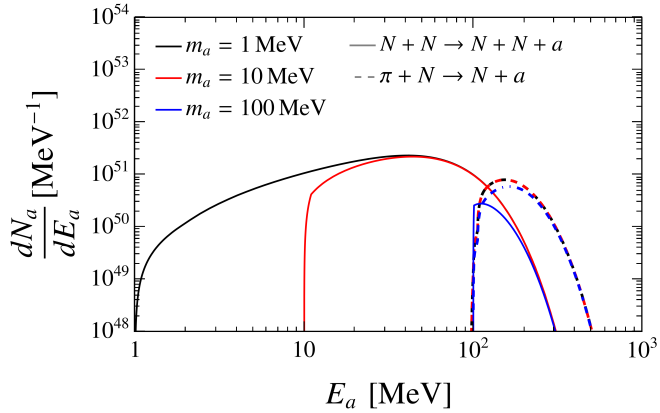

FIG. 2. ALP production spectra via nuclear couplings g_{aN} integrated over the SN volume and over the duration of the burst. Solid lines depict NN bremsstrahlung emission spectra, while pion conversion spectra are displayed as dashed lines. The different colors refer to different ALP masses. In this figure we set the ALP-proton coupling $g_{ap} = 2 \times 10^{-11}$ and $g_{an} = 0$.

$$\frac{dN_{\mathrm{pos}}}{dE_{\mathrm{pos}}} \to BR(a \to e^+e^-) \frac{dN_{\mathrm{pos}}}{dE_{\mathrm{pos}}} , \tag{11}$$

in which $BR(a \to e^+e^-) = \lambda_{\mathrm{ALP}}/\lambda_{\mathrm{ALP},e}$. In this regard, we point out that in the scenario considered the induced photon coupling vanishes around $m_a \simeq 147\,\mathrm{MeV}$ and electronic decays result in the only viable decay channel. On the other hand, around $m_a \simeq 135\,\mathrm{MeV}$ the induced ALP-photon coupling shows a pole, so that the total number of emitted ALPs decay via photon pairs. We refer the reader to Ref. [44] for further discussions on these aspects. As an exemplary case, in the following we will set $g_{ap} = 2 \times 10^{-11}$, which is the maximum value allowed by current astrophysical constraints introduced in Ref. [44].

Sterile neutrinos

MeV-scale sterile neutrinos have interesting phenomenological implications. In the following, we will consider sterile neutrinos with a mass above 10 MeV to safely avoid any possible resonant production and feedback on the SN, which usually occurs below this threshold [13, 76, 77]. In this mass range the mixing of a sterile neutrino with electron neutrino is very constrained [78], therefore we assume that the sterile neutrino is mixed dominantly with one active neutrino ν_α, with $\alpha = \mu, \tau$, such as

$$\begin{aligned} \nu_\alpha &= U_{\alpha 1}\,\nu_\ell + U_{\alpha 4}\,\nu_4 \ , \\ \nu_s &= -U_{\alpha 4}\,\nu_\ell + U_{s4}\,\nu_4 \ , \end{aligned} \tag{12}$$

where ν_ℓ and ν_4 are the light and the heavy mass eigenstates, respectively, U is the unitary mixing matrix, link-